

## William McLean

**AI Developer Advocate & Chief Technology Officer | Google Developer Expert (GDE)**

**AI/Cloud | AWS Community Builder**

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### Professional Summary

Developer Relations leader and CTO who has been **building software agents for 30+ years** — from IBM's agent-oriented OPC Tracker Agent (1992) and a master's thesis on extensible agent technology to production agentic AI systems built on Gemini, Vertex AI, the Model Context Protocol (MCP), and the A2A protocol. **Google Developer Expert (GDE) for AI and Cloud, AWS Community Builder (AI Engineer category), and Google TPU Builders participant**, holding 9 Google Cloud certifications. Published the **first community port of Gemma 4 to AWS Inferentia2 hardware** (Hugging Face). Author of **300+ technical articles** and 170+ open-source repositories spanning multi-agent systems, LLM serving on Cloud TPU and GPUs, and agentic coding tools including Claude Code, OpenAI Codex, and Gemini CLI.

### Core Competencies & Skills

- **Developer Relations & Advocacy:** Developer Tools & SDKs, Community Engagement, Technical Content Creation (Blogs, Code Samples, Demos), Developer Programs, API Design & Review, Developer Feedback Loops, Speaking & Evangelism.
- **Artificial Intelligence (AI) & Machine Learning (ML):** Generative AI (GenAI) — Gemini, Gemma, Anthropic Claude, OpenAI GPT; AI Agents & Agentic Workflows, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vertex AI, Hugging Face, Model Context Protocol (MCP), Agent-to-Agent (A2A) Protocol, Agent Development Kit (ADK), Large Language Model (LLM) Orchestration, LLM Serving & Inference (vLLM, Cloud Tensor Processing Units / TPU v6e, AWS Inferentia2, NVIDIA L4), Model Quantization (QAT), Model Evaluation & Benchmarking, MLOps, TensorFlow, PyTorch, Automated Code Analysis.
- **Agentic Coding & AI Developer Tools:** Anthropic Claude Code (MCP integrations, skills, sub-agents), OpenAI Codex, Gemini CLI, Antigravity CLI, Kiro — building, porting, and benchmarking MCP servers and agent toolchains across all major AI coding platforms.
- **Cloud Architecture:** Google Cloud Platform (GCP Expert), AWS (Community Builder), Azure; Multi-Cloud Strategy, Serverless (Cloud Run, Lambda), Kubernetes (GKE, EKS), Terraform, CI/CD, Infrastructure as Code (IaC).
- **Technical Leadership:** CTO-level Strategy, Board Advising, Investor Relations, Cross-Functional Team Leadership, Strategic Planning, Influence without Authority.
- **Software Engineering:** Distributed Systems, API Design (REST/gRPC), Event-Driven Architecture, Microservices, High-Availability Systems, Client Libraries.
- **Languages:** Python, Rust, Go, TypeScript, Node.js, Java, PHP, C/C++, Bash, REXX.
- **Security & Compliance:** Zero-Trust Architecture, PCI-DSS Compliance (15+ years), PKI, OAuth, Network Security.

### Certifications & Developer Programs

- **Google Developer Expert (GDE):** AI and Cloud (current) — Representing the voice of developer communities, collaborating with Google product managers, and presenting live demos/keynotes globally.
- **TPU Builders (Google):** Deploy and benchmark Gemma-family models on Cloud TPU v6e as part of Google's TPU Builders program; publish TPU deployment guides and open-source MCP tooling for the community.
- **AWS Community Builder (2026):** Selected for the AI Engineer Category; five-time AWS Community Builder Spotlight honoree ([5-Timers Club](#)).
- **dev.to Top Azure Author:** Awarded the dev.to Azure badge, given to the top Azure author each week ([dev.to/xbill](https://dev.to/xbill)).
- **Google Cloud Certified (9x):** Professional Machine Learning Engineer, Generative AI Leader, Cloud Architect, Data Engineer, DevOps Engineer, Cloud Developer, Cloud Security Engineer, Associate Cloud Engineer, Digital Leader.

- **Google Cloud Partner Certifications (2026):** Gemini Enterprise Deployment (secure environment configuration, Google Workspace data integration, Model Armor, search governance) and Gemini Enterprise Agent Development (AI agent development, API integration, prompt tuning, Antigravity IDE).

## Professional Experience

### Developer Advocate | Google Developer Expert (AI/Cloud)

Google Developer Program, New York, NY

2024 - Present

- Represent developer communities to Google product teams through direct feedback channels with Google Cloud and Google DeepMind engineering — reviewing emerging API designs, testing pre-release features, and advocating for developer interests.
- Deploy and benchmark Gemma-family models on Cloud TPU v6e and GPU infrastructure as a **Google TPU Builders** participant, publishing deployment guides and open-source MCP tooling for the community.
- Delivered **10+ keynotes, conference talks, and workshops** in the past year on agentic AI, multimodal agents, and LLM deployment (Build with AI, DevFest, university GDG chapters, hackathons).
- Facilitated hands-on developer workshops as a technical assistant (TA), including a post-event workshop on the **Google Cloud Next 2026** "Agentic Cloud" keynote demos and companion codelabs; sample **currency-agent** (A2A + ADK + MCP) featured in a Google codelab.

### Co-Founder & Chief Technology Officer

Glitnir Ticketing, New York, NY

2002 - Present

- **Developer Enablement & Tooling:** Authored the open-source "**ADK Agent Starter Kit**" and developed high-impact AI utilities, including **mcp-on-cloudrun** (high-performance MCP deployments), to simplify the creation of agentic workflows and client-side implementations for thousands of developers.
- **AI Transformation:** Architected an AI-first engineering culture by integrating agentic coding tools — Claude Code, OpenAI Codex, Gemini CLI — and MCP across the software development lifecycle, from code review and testing to legacy migration.
- **Cloud Infrastructure:** Designed and managed a multi-cloud (GCP & AWS) enterprise SaaS platform processing **\$20M+ in annual transactions** with **99.9% availability**.
- **Regulatory Compliance:** Engineered and maintained a zero-trust architecture, securing **15+ consecutive years of PCI-DSS compliance**.
- **Legacy Modernization:** Developed **refactor**, an AI-assisted toolset for automating the migration of legacy monoliths to modern, cloud-native microservices.

### Chief Technology Officer

Authentidate Sports, New York, NY

2000 - 2002

- Engineered a first-to-market streaming media platform featuring advanced encryption and high-availability failover.
- Directed technical strategy for investor relations and contributed to foundational patent filings for secure media delivery.

### Principal Consultant - Architect

Cap Gemini Ernst & Young, New York, NY

1999 - 2000

- Architected mission-critical trading systems and e-commerce platforms for Fortune 500 clients, including **Reuters, Deutsche Bank, and Chase**.
- Appointed to the elite SBA1 Architecture Council, advising on global architectural standards.

### Chief Designer, OPC Tracker Agent

IBM, Global

1992 - 1999

- Architected Agent-Oriented distributed applications for enterprise-scale systems management across heterogeneous environments.

## Open Source & AI Projects

170+ public repositories at [github.com/xbill9](https://github.com/xbill9). Selected AI-focused work:

- **AI Education: way-back-home** — an immersive AI workshop platform teaching Google Cloud AI agent development (Gemini, Veo, ADK, MCP, A2A) through a narrative rescue mission, used in Build with AI keynotes and workshops.
- **Google Codelab Sample & Starter Kits: currency-agent** family (Python and Rust) — A2A + ADK + MCP sample agent **featured in a Google codelab**, with AG-UI frontend variants; **adk-hello-world** ADK starter kits in Python, Go, and Java; **a2a-inspector** — web-based tool for debugging and validating A2A protocol servers.
- **Multi-Agent Systems (A2A + ADK): multi-agent** — AI Course Creator, a distributed multi-agent system of specialized microservice agents (research, judge, build) orchestrated over the A2A protocol; **a2a-benchmark** and **a2a-multicloud** — cross-language (Python, Go, TypeScript, Rust) and cross-cloud (GCP, AWS, Azure) A2A performance benchmarking suites.
- **Published Models — First Community Port of Gemma 4 to AWS Inferentia2:** Compiled and published the first community Gemma 4 models for AWS Inferentia2 (Neuron) on Hugging Face — **gemma-4-12B-it-inferentia2**, **gemma-4-E4B-it-inferentia2**, and **gemma-4-E2B-it-inferentia2** ([huggingface.co/xbill9](https://huggingface.co/xbill9)) — with companion Docker images at **~1,000 combined pulls** ([hub.docker.com/u/xbill9](https://hub.docker.com/u/xbill9)) and a published field report.
- **TPU & LLM Serving: TPU Sprint MCP Toolkit** (aisprintapr2026) — nine specialized MCP servers that give LLMs "eyes and hands" for deploying and optimizing workloads on Cloud TPU v6e and serverless GPUs (Cloud Run L4), companion to published Gemma-on-TPU deployment guides; **gemma4-tips** — self-hosted Gemma 4-powered DevOps/SRE agents served via vLLM on Cloud Run GPUs, with AWS and Azure variants.
- **Agentic Coding Tool Integrations:** MCP servers and agents built for and ported across **Claude Code, OpenAI Codex, Gemini CLI, Antigravity CLI, and Kiro** — Looker BI (**looker-mcp-claude**), Gemini image and video generation (**nb2lite**, **omni-flash-video-agent**), Google Workspace (**workspace-mcp**), and Google Developer Knowledge (**knowledge-api**); plus production-grade Rust MCP servers (stdio and streaming-HTTP on Cloud Run with Identity-Aware Proxy) and "Cymbal Superstore" MCP implementations across ~19 programming languages.

## Speaking & Community Leadership

- **Speaker | LLM Day NYC | June 2026:** Live demo and talk on **Gemma 4**.
- **Keynote Speaker | Hack Michigan - Way Back Home | May 2026:** Keynote on multimodal, real-time AI agent development.
- **Keynote Speaker | Build with AI - Way Back Home | March 2026:** Keynote on AI agent development and the evolution of the software agent paradigm.
- **Speaker | UofT Build with AI, GDGoC UofT Scarborough | Mar 2026:** Presented **Multimodal ADK with Live Models**.
- **Speaker | Agent Glow Hackathon | Mar 2026:** Presented **Multimodal Gemini Live** and Vertex AI integration.
- **Speaker | Detroit Black History Month Innovation Summit | Feb 2026:** Presented **"LLM Deployment"** on enterprise MLOps and Gemini/Vertex AI integration.
- **Additional Workshops & Mentoring:** Build with AI Workshops (GDG Queens 2026, DevFest Brooklyn/Queens 2025), Future of Data and AI: Agentic AI Conference (2026), Self Improving Agents Hack NYC (2026); Technical Mentor for Agentverse NYC, Accelerate AI with Cloud Run, and Build with AI NYC (2025).

## Education

### Master of Science (M.Sc.), Computer Science

University of Western Ontario

- *Thesis: Distributed Application Management and Extensible Agent Technology*
- *Recipient of the NSERC Canadian Post Graduate Scholarship.*

### Honours Bachelor of Science (B.Sc.), Computer Science

University of Western Ontario

## Publications

300+ technical articles published across [dev.to](#) and [Medium](#). Highlights:

- **"Cross Language A2A Agent Development and Testing"** - Google Cloud Blog
- **"Building AI Agents with the GO Agent Development Kit (ADK)"** - Google Medium
- **"TPU Deployments with Gemma 31B, v6e-8, and Antigravity CLI"** - Medium
- **"Porting Gemma-4 (2B/4B/12B) to AWS Inferentia2"** - dev.to
- **"MCP Configuration for Looker with Claude Code"** - dev.to
- **"Gemma-SRE: Self-Hosted vLLM Infrastructure Agent"** - Medium
- **"Extending a MCP/A2A Currency Agent with AG-UI and Antigravity CLI"** - Medium
- **"Debugging Deployments with Gemma 12B, TPU v6e-1, MCP, and Antigravity CLI"** - Medium
- **"Cross Cloud A2A Agent Benchmarking with Antigravity CLI"** - Medium